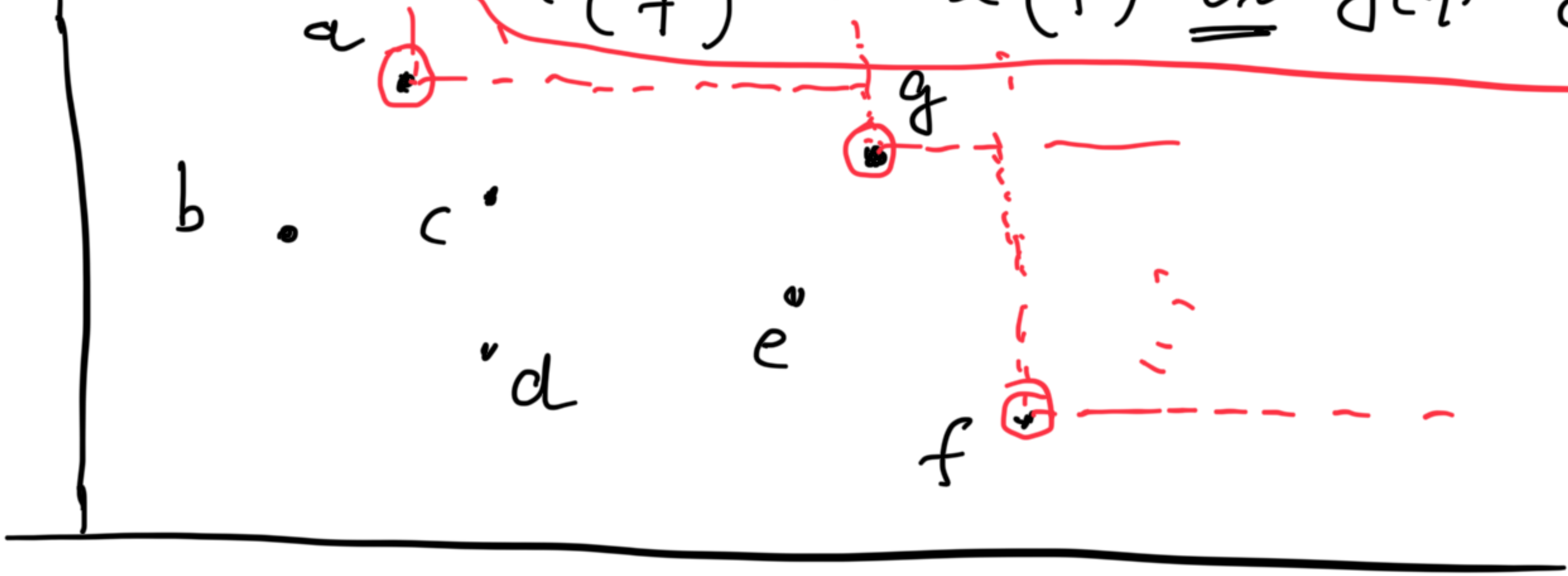
Lecture 5 DAA Feb 4

Defn 1

Given a set S of points $(x_i, y_i)_{1 \le i \le n}$ we call a point $p \in S$ <u>maximal</u> if

$\forall q \in S - \{p\}$

$x(q) < x(p)$ <u>or</u> $y(q) < y(p)$



Defn 2

p <u>dominates</u> q if $x(p) > x(q)$ and $y(p) > y(q)$

$p \succ q$

A point p is maximal if is <u>not</u> dominated by any other $q \in S - \{p\}$

Defn 1 and 2 are equivalent

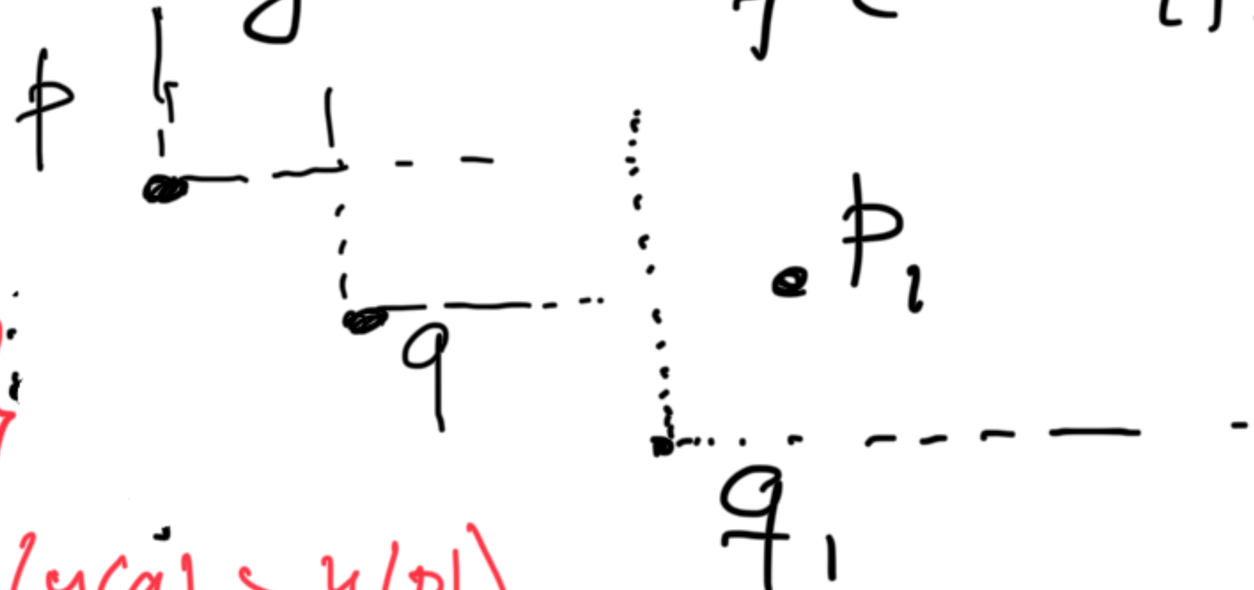
p is maximal if $\forall q$

$q \not\succ p$ (does not dominate).

$\text{Not}\ \{x(q) > x(p) \text{ and } y(q) > y(p)\}$

$\Leftrightarrow \text{Not}\,(x(q) > x(p)) \text{ or } \text{Not}\,(y(q) > y(p))$

$\Leftrightarrow x(q) < x(p) \text{ or } y(q) < y(p).$



Maximal points form a staircase

Given n points, find all maximal points.

Simple/naive $O(n^2)$

Can we do better?

Divide and conquer

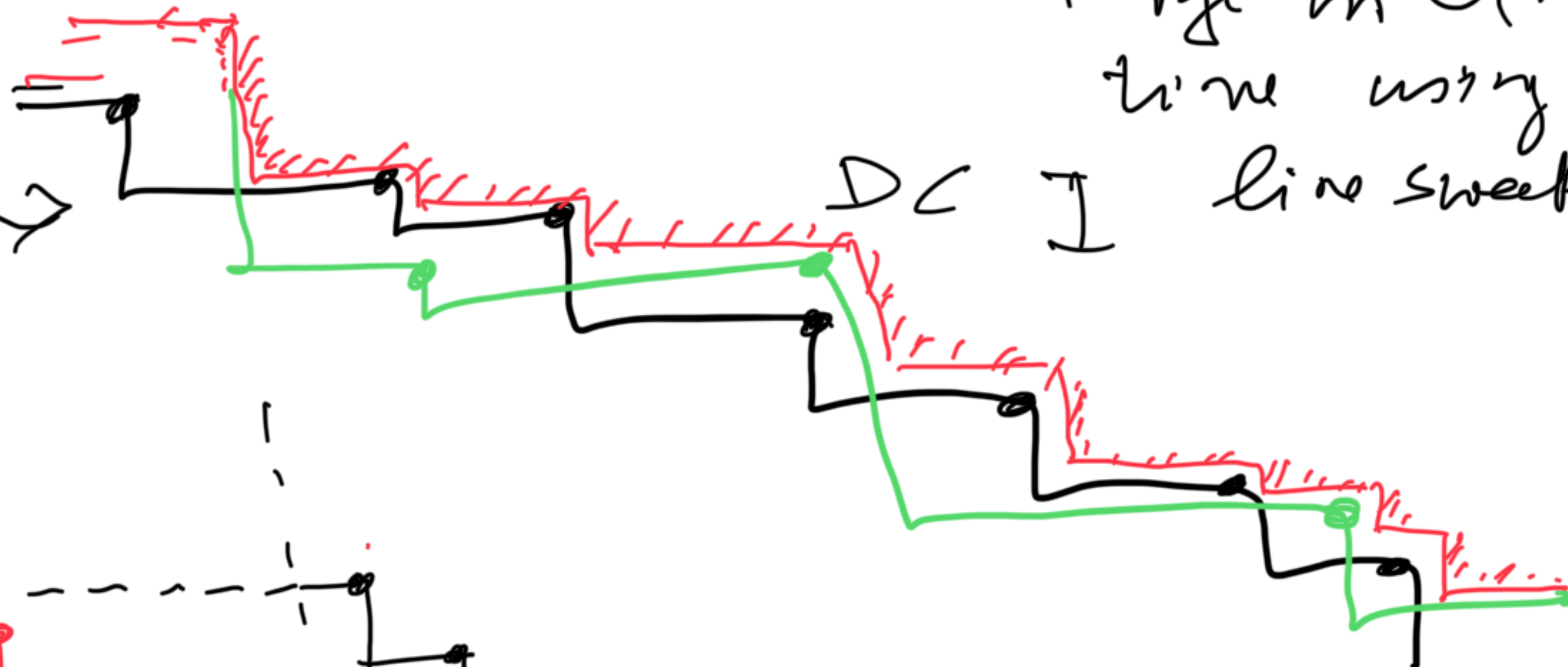
(1) Divide into two (arbitrary) sets of equal size $\sim \frac{n}{2}$

(2) Recursively find maximal points in each

(3) Combine

$T(n) = 2T(\frac{n}{2}) + O(n)$

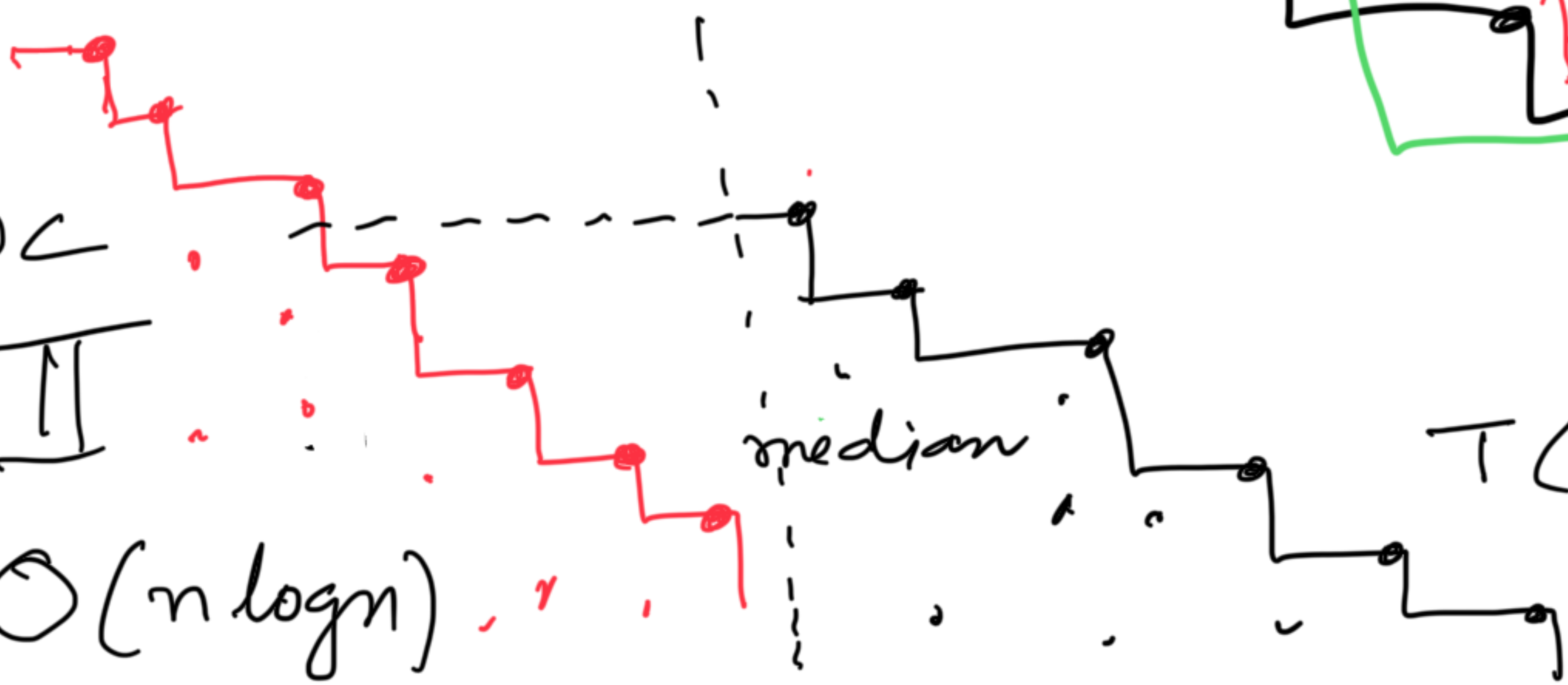
Merge in $O(n)$ time using line sweep



DC I

DC II

$O(n \log n)$



$T(n) = 2T(\frac{n}{2}) + O(n)$

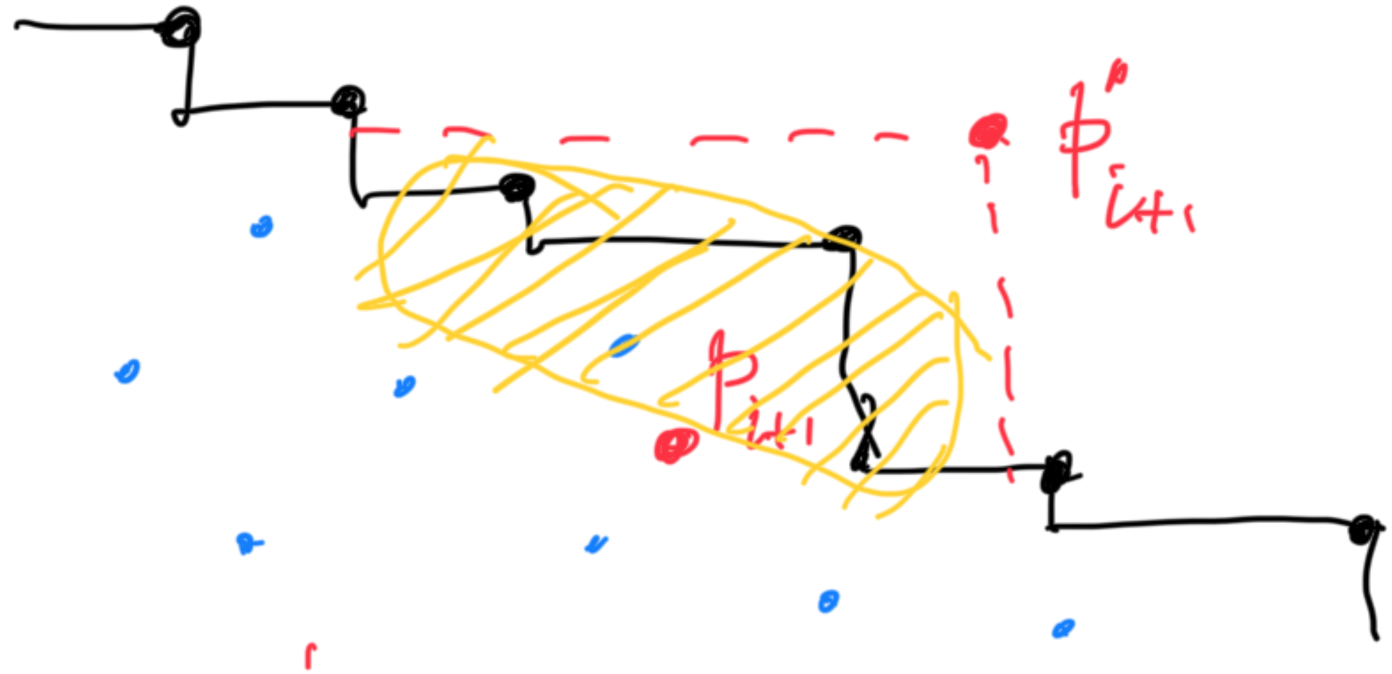
median

Lecture 6 Feb 6

+ O(n) for combining the staircases and eliminating part of it

Incremental construction

At any intermediate stage of i points



above / below staircase test

Use binary searches along X and Y to determine the portion of the staircase that will be erase

Time $= O(\log(i))$ using BBST

$= O(\log n)$

If k_i points are erased in step i then the cost of adjusting the BBST is $O(k_i \log n)$

Total cost of all the n steps

$$= O\left(\sum T_i\right)$$

$$= O\left(\sum_{i=1}^{n} \log n + k_i \log n\right)$$

$$= O(n \log n) + O\left(\sum_{i=1}^{n} k_i \log n\right)$$

$O(n \log n)$ since $\sum k_i \leq n$

Line sweep for maximal

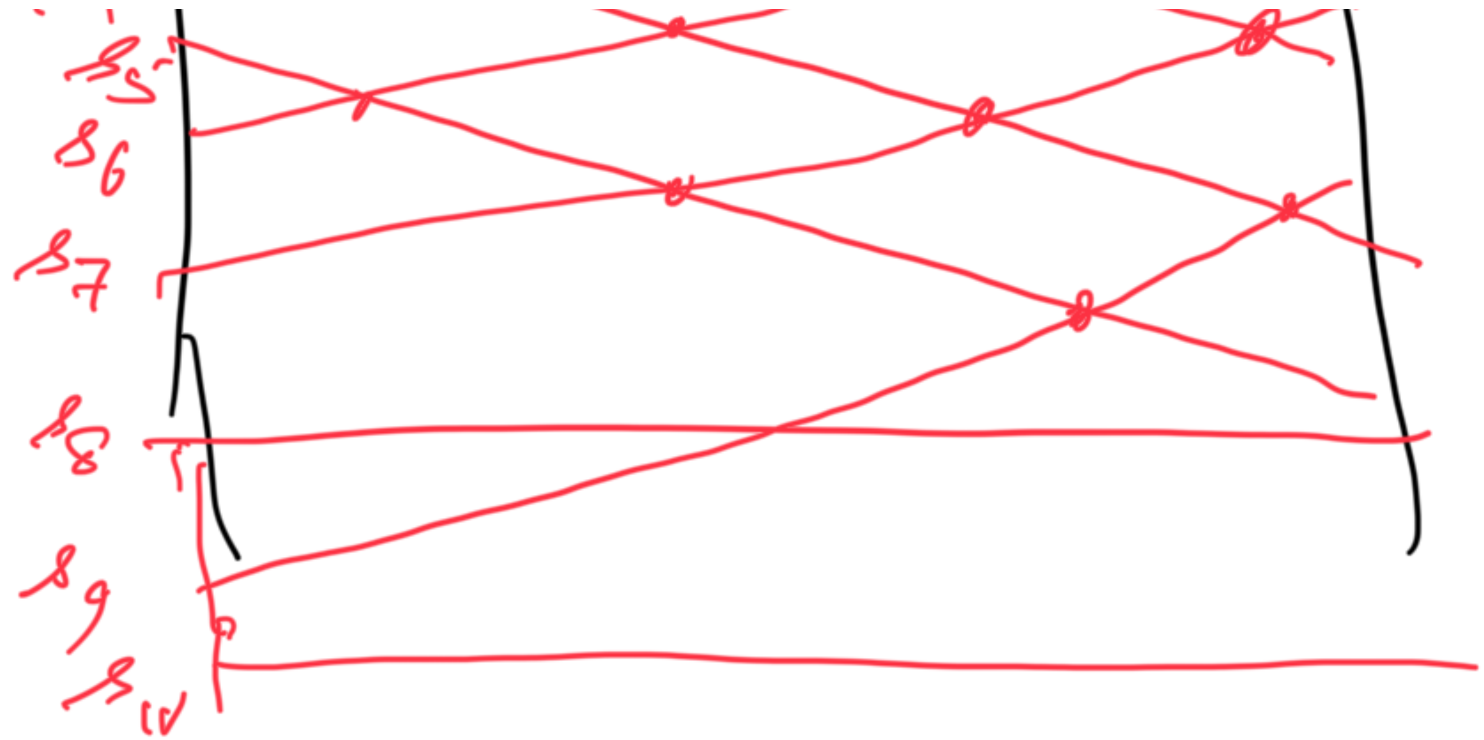
Suppose we start the sweep from the rightmost point (highest x coordinate in the sorted order)

For any point q that hits the line we need to compare $y(q)$ with the highest y coordinate seen thus far (which the last step of the staircase)

Cost of testing whether q is maximal $= O(1)$

$\Rightarrow$ $O(n)$ time for sweeping
$+ \; O(n \log n)$ " for initial sort

$O(n \log n)$ overall



Compute the total number of intersections

Observation: ~~Among~~ Every intersection between segments s_i and s_j flips the ordering of s_i and s_j

$\Rightarrow$ Counting the total no of flips = # intersections.

Design an $O(n\log n)$ algorithm